

Hamza Mumtaz

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EDUCATION & PROFESSIONAL DEVELOPMENT

- Bachelor of Engineering (Electrical Engineering) – Ontario Tech University April 2026
- Minor: Sustainability and Energy Management
- Dassault Systèmes Certification for Mechanical Design
- WHMIS Certified/ Workplace Health and Safety
- RedCross First Aid and CPR

TECHNICAL SKILLS

Electrical & Control Systems Concepts: Electronic Circuit Design, Digital Systems, Microprocessors, Circuit Analysis, Control Systems, Embedded Systems, Signal Conditioning, PID Control, Closed-Loop Control, CAN Communication

Mechanical & System Integration: System Integration, Sensor Integration, Power Distribution, Wiring Harness Design, Mechanical Design, Manufacturing, Hardware Debugging, Test & Validation

Tools: SolidWorks, Multisim, NX, MATLAB, Simulink

Fabrication: Soldering, Wiring, Mechanical Assembly, Brazing, MIG/Stick welding, Coatings & Finishing

RELATED PROFESSIONAL EXPERIENCE

Mechanical Engineer/Race Systems Analyst, Mu Engineering, September 2024– Present

- Designed and integrated mechanical, electrical, and control systems for an in-house race car platform, spanning drivetrain, suspension, and aerodynamic components.
- Developed and validated active dynamic sub-systems using telemetry, sensor integration, and control algorithms to improve vehicle handling and response.
- Led CAD modeling, component selection, prototype fabrication, balancing performance requirements, manufacturability, and system integration.
- Conducted simulation-driven and real-world testing to analyze vehicle behavior, validate design assumptions, and iteratively refine system performance.
- Managed iterative design cycles, technical documentation, and project timelines to support structured engineering decision-making.

Engineering Contractor, Mu Engineering, January 2021– August 2024

- Delivered project-based engineering support across diverse technical initiatives with varying mechanical, electrical, and system integration requirements.
- Contributed to system design, CAD modeling, wiring, and prototype fabrication based on project-specific constraints and objectives.
- Supported component and sensor selection, integrating electrical and mechanical considerations into proposed system solutions.
- Prepared technical documentation and design summaries to communicate engineering rationale and proposed solutions to stakeholders.

Electrical Apprentice (Electrician), Eam Mosca, December 2022 – May 2023

- Assembled and wired electrical panels and junction boxes from engineered schematics, ensuring compliance with electrical and safety standards.
- Installed and wired electrical components including motors, transformers, and pushbutton interfaces according to design drawings and codes.
- Assembled Allen-Bradley PLCs and in-house HMI systems for industrial control applications.
- Diagnosed and corrected wiring and component issues identified during assembly and testing.

ADDITIONAL WORK EXPERIENCE

Quality Control & Inspection, General Motors, January 2022 – April 2022

- Performed quality inspections on completed GM vehicles to identify defects and deviations from manufacturing standards.
- Documented inspection findings using internal diagnostic and reporting systems to support defect tracking and resolution.
- Maintained a high level of accuracy and consistency while working within a production-focused environment.

Operations & Safety Coordinator, Kenny U-Pull, May 2024 – Present

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- Conducted workplace safety inspections, hazard assessments, and regulatory compliance checks in accordance with occupational health and safety standards.
- Led safety briefings and toolbox talks, investigated incidents, and implemented corrective actions to prevent recurrence.
- Ensured readiness and compliance of safety equipment, including fire extinguishers and eyewash stations.
- Acted as a liaison between employees and management to address safety concerns and promote safe work practices.

Automotive Mechanic, Kenny U-Pull, May 2024 – Present

- Performed systematic vehicle decommissioning, including complete fluid drainage and safe handling of hazardous materials.
- Disassembled and removed mechanical and drivetrain components using proper tools and procedures to support recycling and part recovery.

Customer Service Representative, Great Hobbies, April 2022 – July 2022

- Diagnosed and repaired R/C and electronic systems, including fault identification, soldering, and component replacement.
- Provided technical support in a fast-paced environment, communicating repair findings and solutions clearly to customers.

Electronics Repairs Specialist (self-employed), January 2014– January 2021

- Diagnosed and repaired consumer electronic systems, performing fault isolation, component-level troubleshooting, and soldering.
- Designed and modified electronic circuits to restore functionality or improve reliability based on application requirements.

SELECTED PROJECTS

Autonomous Electric Scooter – Capstone Project

- Designed and integrated a CAN-based distributed control architecture for an autonomous electric scooter, supporting sensor fusion, actuator control, and inter-module communication.
- Led sensor and actuator integration, developing CAN addressing, messaging, and interface logic while coordinating system integration with the software autonomy stack.
- Designed a transferable embedded controller platform integrating a microcontroller, CAN termination, motor drivers, power management, and actuator interfaces, supporting system-level testing and validation.

JTCC-Inspired Honda Accord – Vehicle Control & Electronics Platform

- Developed a vehicle-scale electronics and control platform integrating embedded controllers, sensor networks, and telemetry to support real-time vehicle monitoring and experimentation.
- Designed and implemented control, data acquisition, and signal-conditioning subsystems, supporting system-level testing, validation, and iterative refinement.
- Architected distributed telemetry and IoT concepts using embedded microcontroller nodes to evaluate scalable data sharing and modular vehicle electronics architectures.

Active Aero Control System – 1/10 Scale Vehicle

- Designed and implemented a closed-loop active aerodynamic system using dual IMUs and PID control to dynamically modulate aerodynamic surfaces during braking and cornering.
- Developed embedded firmware to perform real-time sensor processing, control computation, and servo actuation on a microcontroller platform.
- Validated system performance through physical testing, tuning control gains, and analyzing vehicle behavior during dynamic maneuvers.

Accessibility Keyboard – Assistive Technology Design Project

- Designed and built a custom accessibility-focused keyboard incorporating electronic and mechanical modifications to improve usability under constrained input scenarios.
- Prototyped and validated the device through functional testing and iterative refinement, evaluating ergonomics, reliability, and user interaction.
- Applied human-centered design principles and structured project planning to balance accessibility requirements, manufacturability, and system robustness.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- VP Ontario Tech Car Club (2023–2025)
- OTU Drone Design Lead (2019–2020)
- First Robotics Competition (2015–2018)